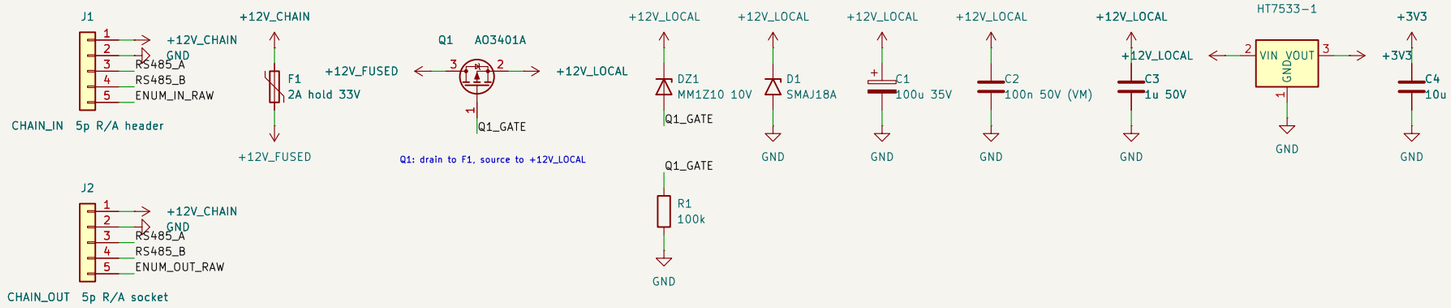
League-table module board v0.1 — per design/DESIGN-1.md — generated by scripts/gen_schematic.py, do not hand-edit

One board per module (digit and name). Single sheet; power symbols are global rails, labels are sheet-local.

A CHAIN + INPUT PROTECTION

J1 = right edge (toward the trunk), male right-angle header. J2 = left edge, female socket: the live end of a row is shrouded.
+12V_CHAIN and GND pass straight through J1 -> J2 on copper. Only the local branch F1 -> Q1 -> +12V_LOCAL is protected.



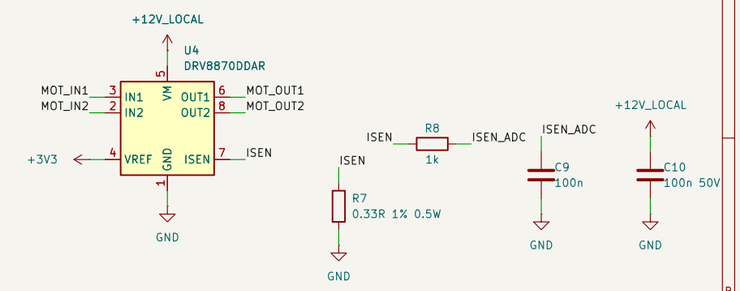
Pinout: 1 +12V 2 GND 3 A 4 B 5 ENUM

B 3.3 V LOGIC RAIL HT7533-1 (30 V in, 100 mA)

HT75xx-1-SOT89: 1 GND 2 VIN (tab) 3 VOUT. Check Holtek's datasheet at layout.

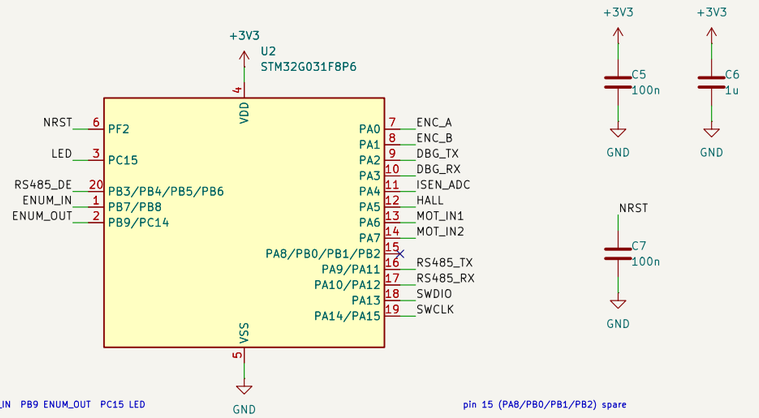
E MOTOR DRIVER DRV8870DDA

I_TRIP = VREF / (10 x R7) = 3.3 / (10 x 0.33) = 1.0 A. ISEN -> R8/C9 -> ADC for jam detection.
IN1/IN2 low (internal pull-downs) = coast while the MCU is in reset.



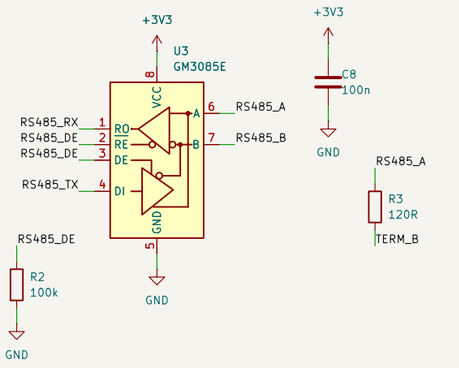
C MCU STM32G031F8P6 (64 KB)

Pin map checked against ST STM32_open_pin_data (STM32G031F(4-6-8)Px, GPIO-STM32G03x modes) 2026-10-15.
TIM2 encoder (PA0/PA1 AF2) TIM3 PWM (PA6/PA7 AF1) USART1 TX/RX PA9/PA10 (remap, AF1) + DE PB3 (AF4) USART2 debug PA2/PA3 (AF1)
Co-bonded pads stay analog in firmware: PB8 (1), PC14 (2), PA8/PB0/PB1 (15), PA11/PA12 (16/17), PA15 (19), PB4/PB5/PB6 (20).

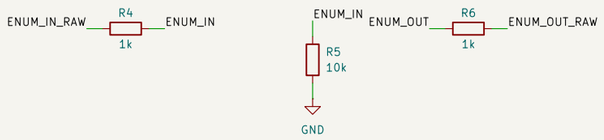


D RS-485 GM3085E (MAX3485 pinout; project symbol ltm:GM3085E)

DE and /RE tied: R2 holds DE low through reset. Termination: trunk ends only (JP1 for bench use). Bias resistors at the host only.

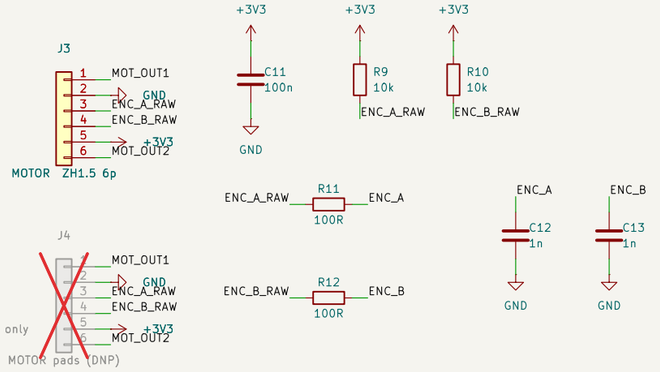


ENUMERATION 1k series both ways (hot-plug into an unpowered MCU); 10k pull-down on ENUM_IN



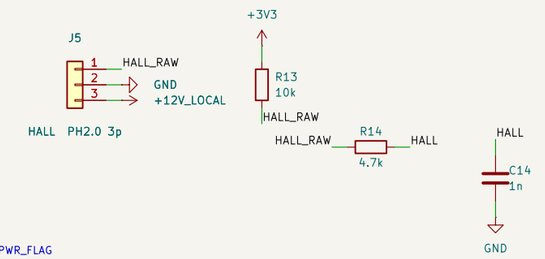
F MOTOR + ENCODER N20 (Waveshare DCGM-N20-12V-EN-200RPM)

Pin 1 = M1: order M1 GND C1 C2 VCC M2 (web listings; Drew's guess). Check with a meter at bring-up: reversed puts 3.3 V on the encoder's GND.
J4 = solder pads for the bare Waveshare lead (DNP), same nets as J3.



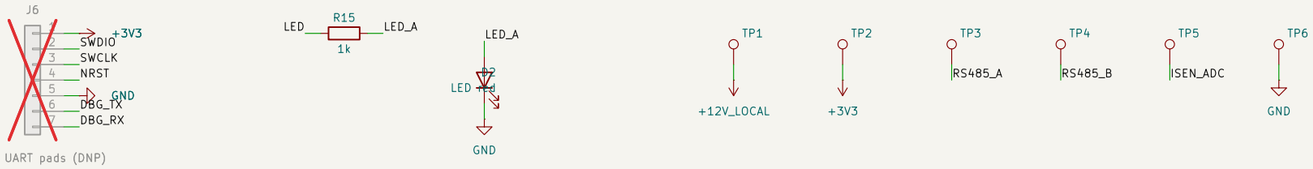
G HALL A3144 (dowjames v5.1 connector and pinout)

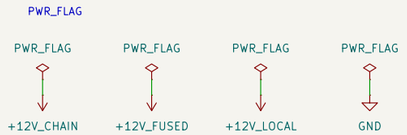
1 OUT 2 GND 3 +12V. Open collector: pulled up to 3.3 V here. R14 4.7k limits current if 12 V ever reaches OUT.



H DEBUG / LED / TEST POINTS

J6: SWD (first flash) + USART2 debug. Pads only (DNP); a pogo jig or a hand-soldered header.





Per design/DESIGN-1.md; generated by scripts/gen_schematic.py
Split-flap league table

Sheet: /
File: league-table-module-v0.1.kicad_sch

Title: League-table module board v0.1

Size: A3	Date: 2026-10-15	**Rev: 0.1**
KiCad E.D.A. 10.0.5		Id: 1/1